

A new well-balanced ADER discontinuous Galerkin method for the shallow water flows in channels with irregular geometry

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Abstract

This article develops a new well-balanced discontinuous Galerkin method for the shallow water flows in channels with irregular geometry and over uneven bottom topography, which preserves the still water steady state solutions. Different from the traditional discontinuous Galerkin methods using the multi-stage Runge-Kutta approach, the current method applies the ADER (Arbitrary DERivatives for space and time) approach for the temporal discretization. To achieve the local time evolution, the numerical solutions, which are in the forms of truncated spatiotemporal Taylor polynomials, apply the differential transformation procedure to obtain spatiotemporal derivatives through the spatial ones. The differential transformation procedure is more efficient and the coding is more concise than the Cauchy-Kowalewski procedure from the traditional ADER methods. Furthermore, a combination of well-balanced numerical fluxes and simple source term discretizations leads to well-balanced property for the still water steady state. In summary, the proposed method is one-step, one-stage, fully-discrete, and theoretically achieves arbitrary high-order accuracy in both space and time. Additionally, the current method needs less computer memory storage compared with the traditional Runge-Kutta discontinuous Galerkin methods. Extensive numerical results indicate that the current method maintains the well-balanced property, enjoys high-order accuracy and keeps steep discontinuity transition. Moreover, the resulting method resolves well small perturbations to the still water steady state even on relatively coarse

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mesh.

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1. Introduction

The shallow water flows in channels are widely used to describe the shallow water fluids in channels, where the channel geometry is characterized by variable channel breadths along with uneven bottom topographies. Numerical simulation to the shallow water flows in channels is a very popular research approach in the fields of computational hydrodynamics. This article presents a new discontinuous Galerkin (DG) method for the shallow water flows with variable channel breadths and over uneven bottom topographies, which enjoy the following form [1, 2]:

$$\begin{aligned} H_t + Q_x &= 0, \\ Q_t + \left(\frac{Q^2}{H} + \frac{1}{2}g\sigma h^2 \right)_x &= \frac{1}{2}gh^2\sigma_x - g\sigma hb_x, \end{aligned} \tag{1}$$

where $\sigma(x)$ is the channel breadth, $b(x)$ denotes the bottom topography, $h(x, t)$ stands for the water height, $u(x, t)$ is the fluid velocity, $H(x, t) = \sigma h$ is the wetted cross-sectional area, $Q = Hu$ is the flow rate of discharge, and g is the gravitational constant. In particular, when the channel breadth function $\sigma(x)$ is a constant, the above system (1) reduces to the traditional shallow water equations, i.e.,

$$\begin{aligned} h_t + (hu)_x &= 0, \\ (hu)_t + \left(hu^2 + \frac{1}{2}gh^2 \right)_x &= -ghb_x. \end{aligned} \tag{2}$$

The above nonlinear model (1) along with the source term belongs to the class of hyperbolic balance laws, and maintains various non-trivial equilibrium solutions (the steady-state solutions), which leads to delicate balance between the flux gradient and the source term. This study concentrates on a special still water steady state with zero velocity, namely

$$u = 0, \quad h + b = C, \tag{3}$$

with constant C .

Since the nonlinear system (1) mathematically maintains the steady state solutions, it is desirable for numerical methods to maintain the steady state solutions up to the machine precision. However, standard numerical methods, which are accompanied by simple numerical fluxes along with direct source term discretizations, generally fail to maintain the steady state solutions in the discrete state. In addition, standard numerical methods often occur oscillations near strong discontinuities. Moreover, even though on very fine meshes, non-physical oscillations will not disappear automatically. There are two main reasons for this failure: on one hand, the numerical flux can not be consistent with the physical flux; on the other hand, the discretizations of the flux gradient and the source term can not accurately offset each other. To overcome the disadvantages of the traditional methods, Greenberg et al. [3, 4] have originally developed a kind of methods, which can preserve the steady state until the machine accuracy and are named as the well-balanced (W-B) methods. It is worth noting that even on relatively rough meshes, the W-B methods can capture small disturbances to the steady state solutions and greatly save the computational cost [5]. In view of the good performance of the W-B methods, researchers have developed a number of W-B methods for shallow water equations (2), see [6–19] and the references cited therein.

Concerning the W-B methods for the system (1), representative study results mainly include: Garcia-Navarro and Vázquez-Cédon [20] have developed the Roe’s discretization with the flux difference splitting. Balbás and Karni [21] have constructed second-order central method with respect to rectangular channels. Hernández-Duenas et al. [2, 22] have extended this method to arbitrary type of cross section. Murillo and García-Navarro [23] have carried out the research of the energy balanced property. Hernández-Duenas and Beljadid [24] have built a new semi-discrete non-oscillatory central-upwind method by means of artificial viscosity. Xing [25] has proposed finite volume weighted essentially non-oscillatory (WENO) schemes based on the hydrostatic reconstruction technique. Qian et al. [26] have designed Runge-Kutta DG (RKDG) method on the basis of an extrapolation on the source term approximation. Recently, Wang et al. [27] have established a finite difference WENO scheme using a flux modification technique. More recently, Dong and Li [28] have constructed a non-

staggered central scheme. Liu [29] has considered the finite volume schemes with a global flux term.

In fact, the traditional DG methods take the discontinuous piecewise polynomial approximation spaces for both the trial solution and the test function [30], and are within the framework of semi-discrete methods. In general, the resulting semi-discrete DG methods apply the multi-stage Runge-Kutta approach to realize the temporal discretization [31–44] and are named as the RKDG methods. In short, the RKDG methods are actually multi-stage methods.

For the temporal discretization, besides to the traditional Runge-Kutta approach, researchers have attempted different explorations in the literature. Qiu et al. [45] have developed a one-stage DG method taking the Lax-Wendroff approach for the temporal discretization. In addition, the ADER method belongs to another category of one-step, one-stage, fully-discrete method [46–48], which achieves arbitrary high-order accuracy in both space and time. The key ingredient of the ADER method is to replace the time derivatives by the space derivatives using the well known Cauchy-Kowalewski (C-K) procedure (also termed as the Lax-Wendroff procedure in the literature) [49, 50]. Recently, Dumbser et al. [51, 52], Duan and Tang [53] have applied the local continuous space-time Galerkin predictor approach [54] to replace the C-K procedure in the framework of the DG methods. Dumbser and Munz have presented a new DG method with the one-stage ADER approach for the temporal discretization, which needs to solve generalized Riemann problems at inter-cells [55–62]. However, due to the complicated flux gradient and the presence of the source term, the C-K procedure will become more and more cumbersome. Actually, Pukhov [63] and Zhou [64] have proposed a differential transformation (DT) procedure for the nonlinear initial value problems. Then, the DT procedure is extended to multi-dimensional system [65–67]. Subsequently, Norman and Finkel [68] have developed a multi-moment finite volume scheme using the DT procedure to replace the C-K procedure. In addition, Norman and Finkel have found that the DT procedure is more concise and more efficient than the C-K procedure [68]. More recently, Li et al. have developed a high-order W-B ADER-DG method for shallow

water equations over uneven bottom using the DT procedure [69].

The key objective of this article is to build a high-order W-B DG method for shallow water flows in channels with irregular geometry and over uneven bottom (1). But unlike the traditional RKDG methods, the current DG method uses the ADER approach for the temporal discretization and is referred to as the ADER-DG method accordingly. In addition, the proposed method applies the DT procedure [63–67] instead of the C-K procedure to obtain the spatiotemporal derivatives based on the spatial ones. To achieve the W-B property for the still water steady state (3), W-B numerical fluxes are constructed using the hydrostatic reconstruction idea [70, 71] along with a simple source term discretization. In summary, the proposed ADER-DG method is one-step, one-stage, and fully-discrete. Theoretically, the current method can achieve arbitrary high-order accuracy in both space and time. Numerically, the resulting ADER-DG method can save a lot of computer memory compared with the RKDG method with the same order. To our best knowledge, this is the first attempt to apply the ADER-DG method along with the DT procedure to solve the shallow water flows in channels (1).

The structure of this article is organized as follows. Section 2 presents specific construction steps of the W-B ADER-DG method using the DT procedure. The section 3 conducts extensive numerical examples to validate the performance of the proposed method. Finally, some conclusions are drawn in section 4.

2. Construction of the W-B ADER-DG method

2.1. Notations

Firstly, the spatial interval $[a, b]$ is divided into N cells $I_j = [x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}]$ with $x_j = \frac{1}{2}(x_{j-\frac{1}{2}} + x_{j+\frac{1}{2}})$ as the cell center for $j = 1, 2, \dots, N$. The notation $\tau_j = x_{j+\frac{1}{2}} - x_{j-\frac{1}{2}}$ denotes the size of the j -th cell along with $\tau = \max_{1 \leq j \leq N} \tau_j$ being the maximum mesh size. Herein, the notation $\Omega_j = I_j \times [t^n, t^{n+1}]$ stands for the space-time control volume. The approximation function space is defined as follows

$$V_\tau^k = \left\{ v(x) : v(x)|_{I_j} \in P^k(I_j), j = 1, 2, \dots, N \right\}. \quad (4)$$

2.2. General formulation of ADER-DG method with DT procedure

For the convenience of the DT procedure, the original system (1) is reformulated as follows

$$\begin{aligned} H_t + Q_x &= 0, \\ Q_t + \left(\frac{Q^2}{H} + \frac{1}{2}g \frac{H^2}{\sigma} \right)_x &= -\frac{1}{2}gH^2 \left(\frac{1}{\sigma} \right)_x - gHb_x, \end{aligned} \quad (5)$$

or in the below compact form

$$\frac{\partial U(x, t)}{\partial t} + \frac{\partial F(U, \sigma)}{\partial x} = S(x, t), \quad (6)$$

where

$$U = (H, Q)^\top, \quad F(U, \sigma) = \left(Q, \frac{Q^2}{H} + \frac{1}{2}g \frac{H^2}{\sigma} \right)^\top, \quad S = \left(0, -\frac{1}{2}gH^2 \left(\frac{1}{\sigma} \right)_x - gHb_x \right)^\top$$

denote the conservative variables, the physical flux and the source term, respectively.

It should be noted that the DG method here is different from the DG method in [26], which uses the multi-stage Runge-Kutta approach for the temporal discretization. In this study, the proposed DG method applies the ADER approach for the temporal evolution and then is one-stage, fully-discrete accordingly.

Firstly, multiplying the system (6) with an arbitrary test function $v(x) \in V_\tau^k$ and integrating over a given space-time control volume Ω_j , then yields the following weak form

$$\begin{aligned} \int_{I_j} U_\tau(x, t^{n+1})v(x) dx - \int_{I_j} U_\tau(x, t^n)v(x) dx - \iint_{\Omega_j} F(U_\tau(x, t), \sigma_\tau(x))v_x dx dt \\ + \widehat{F}_{j+\frac{1}{2}}^- \cdot v \left(x_{j+\frac{1}{2}}^- \right) - \widehat{F}_{j-\frac{1}{2}}^+ \cdot v \left(x_{j-\frac{1}{2}}^+ \right) = \iint_{\Omega_j} S_\tau(x, t)v(x) dx dt, \quad \text{for } j = 1, 2, \dots, N, \end{aligned} \quad (7)$$

where $\widehat{F}_{j-\frac{1}{2}}^+$ and $\widehat{F}_{j+\frac{1}{2}}^-$ are numerical fluxes at inter-cells. For example, the Lax-Friedrichs numerical fluxes hold the below simple forms

$$\begin{aligned} \widehat{F}_{j-\frac{1}{2}}^+ &= \frac{1}{2} \int_{t^n}^{t^{n+1}} \left[F \left(U_{\tau, j-\frac{1}{2}}^-(t), \sigma_{\tau, j-\frac{1}{2}}^- \right) + F \left(U_{\tau, j-\frac{1}{2}}^+(t), \sigma_{\tau, j-\frac{1}{2}}^+ \right) - \alpha \left(U_{\tau, j-\frac{1}{2}}^+(t) - U_{\tau, j-\frac{1}{2}}^-(t) \right) \right] dt, \\ \widehat{F}_{j+\frac{1}{2}}^- &= \frac{1}{2} \int_{t^n}^{t^{n+1}} \left[F \left(U_{\tau, j+\frac{1}{2}}^-(t), \sigma_{\tau, j+\frac{1}{2}}^- \right) + F \left(U_{\tau, j+\frac{1}{2}}^+(t), \sigma_{\tau, j+\frac{1}{2}}^+ \right) - \alpha \left(U_{\tau, j+\frac{1}{2}}^+(t) - U_{\tau, j+\frac{1}{2}}^-(t) \right) \right] dt, \end{aligned} \quad (8)$$

where $\alpha = \max(|u| + \sqrt{gh})$ denotes the largest eigenvalues of the Jacobian matrix $\frac{\partial F}{\partial U}$ and is used to estimate the wave propagation speed on the whole spatial domain at t^n .

2.3. The DT procedure for the shallow water flows in open channels

The first step is to implement the initialization and then to obtain

$$\begin{aligned} U_\tau(x, 0) &= (H_\tau, Q_\tau)^\top(x, 0) = \sum_{k_x=0}^k \tilde{U}(k_x, 0)(x - x_j)^{k_x}, \\ b_\tau(x) &= \sum_{k_x=0}^k \tilde{b}(k_x)(x - x_j)^{k_x}, \\ \sigma_\tau(x) &= \sum_{k_x=0}^k \tilde{\sigma}(k_x)(x - x_j)^{k_x}, \\ \left(\frac{1}{\sigma}\right)_\tau(x) &= \sum_{k_x=0}^k \left(\frac{1}{\sigma}\right)(k_x)(x - x_j)^{k_x}, \end{aligned} \tag{9}$$

approximating $U(x, 0)$, $b(x)$, $\sigma(x)$ and $\left(\frac{1}{\sigma}\right)(x)$ in each cell I_j with the aid of the L^2 projection.

Then, to obtain high-order temporal accuracy, a local time evolution inside each local space-time control volume Ω_j needs to be realized. Suppose that the numerical solution in cell I_j at time t^n is known with the following form

$$U_\tau(x, t^n) = (H_\tau, Q_\tau)^\top(x, t^n) = \sum_{k_x=0}^k \tilde{U}(k_x, 0)(x - x_j)^{k_x}. \tag{10}$$

Then, to realize the local time evolution, the DT procedure instead of the C-K procedure from traditional ADER methods [47, 48] is applied. The DT procedure was originally proposed for nonlinear initial value problems in the electric circuit analysis [63–67]. Suppose that a function $u(x, t)$ in cell I_j at time t^n is known, the detailed DT procedure is defined as follows

$$\tilde{u}(k_x, k_t) = \frac{1}{k_x!k_t!} \left. \frac{\partial^{k_x+k_t} u(x, t)}{\partial x^{k_x} \partial t^{k_t}} \right|_{x=x_j, t=t^n}, \quad \text{with } u(x, t) = \sum_{k_t=0}^k \sum_{k_x=0}^{k-k_t} \tilde{u}(k_x, k_t)(x - x_j)^{k_x}(t - t^n)^{k_t},$$

where $u(x, t)$ is the original function and $\tilde{u}(k_x, k_t)$ denotes the transformed function. In fact, the transformed function $\tilde{u}(k_x, k_t)$ denotes the expansion coefficient of the truncated Taylor series. Table 1 shows some transformed functions used in this study.

Table 1: Transformed functions of some functions encountered in this study.

Functional form	Transformed form
$q(x, t) = c \cdot u(x, t)$	$\tilde{q}(k_x, k_t) = c \cdot \tilde{u}(k_x, k_t)$
$q(x, t) = \partial_x u(x, t)$	$\tilde{q}(k_x, k_t) = (k_x + 1)\tilde{u}(k_x + 1, k_t)$
$q(x, t) = \partial_t u(x, t)$	$\tilde{q}(k_x, k_t) = (k_t + 1)\tilde{u}(k_x, k_t + 1)$
$q(x, t) = u(x, t)v(x, t)$	$\tilde{q}(k_x, k_t) = \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \tilde{u}(r, s)\tilde{v}(k_x - r, k_t - s)$
$q(x, t) = \frac{1}{u(x, t)}$	$\tilde{q}(0, 0) = \frac{1}{\tilde{u}(0, 0)}$ $\tilde{q}(k_x, k_t) = -\frac{1}{\tilde{u}(0, 0)} \sum_{r=0}^{k_x} \sum_{\substack{s=0 \\ s+r>0}}^{k_t} \tilde{u}(r, s)\tilde{q}(k_x - r, k_t - s)$

Then, applying the DT procedure on both ends of (6) in a componentwise means yields the following recurrence relations for the transformed functions

$$\tilde{U}(k_x, k_t + 1) = -\frac{k_x + 1}{k_t + 1} \tilde{F}(k_x + 1, k_t) + \frac{1}{k_t + 1} \tilde{S}(k_x, k_t), \quad (11)$$

where

$$\begin{aligned} \tilde{F}(k_x + 1, k_t) &= \begin{pmatrix} \tilde{Q}(k_x + 1, k_t) \\ \tilde{G}_1(k_x + 1, k_t) + \tilde{G}_2(k_x + 1, k_t) \end{pmatrix}, \\ \tilde{S}(k_x, k_t) &= \begin{pmatrix} 0 \\ -\tilde{G}_3(k_x, k_t) - \tilde{G}_4(k_x, k_t) \end{pmatrix}, \end{aligned} \quad (12)$$

along with the following intermediate variables

$$\begin{aligned} \tilde{G}_1(k_x, k_t) &= \frac{\tilde{Q}^2}{H}(k_x, k_t) = \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \tilde{Q}^2(r, s) \frac{1}{H}(k_x - r, k_t - s), \\ \tilde{G}_2(k_x, k_t) &= \frac{\widetilde{H^2}}{\sigma}(k_x, k_t) = \sum_{r=0}^{k_x} \widetilde{H^2}(r, k_t) \left(\frac{1}{\sigma} \right) (k_x - r), \\ \tilde{G}_3(k_x, k_t) &= \left[\widetilde{H^2 \left(\frac{1}{\sigma} \right)}_x \right] (k_x, k_t) = \sum_{r=0}^{k_x} \widetilde{H^2}(r, k_t) \left(\frac{1}{\sigma} \right)_x (k_x - r), \\ \tilde{G}_4(k_x, k_t) &= \widetilde{Hb_x}(k_x, k_t) = \sum_{r=0}^{k_x} \tilde{H}(r, k_t) \tilde{b}_x(k_x - r), \end{aligned}$$

and

$$\begin{aligned}
\widetilde{Q}^2(k_x, k_t) &= \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \widetilde{Q}(r, s) \widetilde{Q}(k_x - r, k_t - s), \\
\left(\widetilde{\frac{1}{H}}\right)(k_x, k_t) &= -\frac{1}{\widetilde{H}(0, 0)} \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \widetilde{H}(r, s) \left(\widetilde{\frac{1}{H}}\right)(k_x - r, k_t - s), \\
\widetilde{H}^2(k_x, k_t) &= \sum_{r=0}^{k_x} \sum_{s=0}^{k_t} \widetilde{H}(r, s) \widetilde{H}(k_x - r, k_t - s), \\
\left(\widetilde{\frac{1}{\sigma}}\right)(k_x) &= -\frac{1}{\widetilde{\sigma}(0)} \sum_{r=0}^{k_x} \widetilde{\sigma}(r) \left(\widetilde{\frac{1}{\sigma}}\right)(k_x - r), \\
\left(\widetilde{\frac{1}{\sigma}}\right)_x(k_x) &= (k_x + 1) \left(\widetilde{\frac{1}{\sigma}}\right)(k_x + 1), \\
\widetilde{b}_x(k_x) &= (k_x + 1) \widetilde{b}(k_x + 1).
\end{aligned}$$

For convenience, the detailed steps of the DT procedure are shown in Appendix A.

Starting from the spatial expansion coefficients $\widetilde{U}(k_x, 0)$ in (9) and using the recurrence formula (11) as well as (12) repeatedly give the spatiotemporal expansion coefficients

$$\widetilde{U}(k_x, k_t), \quad \widetilde{F}(k_x, k_t), \quad \text{and} \quad \widetilde{S}(k_x, k_t), \quad \text{for } k_x = 0, 1, 2, \dots, k; \quad k_t = 0, 1, 2, \dots, k - k_x.$$

Once the spatiotemporal expansion coefficients are obtained, the space-time polynomials of degree k enjoy the below forms

$$\begin{aligned}
U_\tau(x, t) &= \sum_{k_x=0}^k \sum_{k_t=0}^{k-k_x} \widetilde{U}(k_x, k_t) (x - x_j)^{k_x} (t - t^n)^{k_t}, \\
F_\tau(x, t) &= \sum_{k_x=0}^k \sum_{k_t=0}^{k-k_x} \widetilde{F}(k_x, k_t) (x - x_j)^{k_x} (t - t^n)^{k_t}, \quad (x, t) \in \Omega_j, \quad \text{for } j = 1, 2, \dots, N. \\
S_\tau(x, t) &= \sum_{k_x=0}^k \sum_{k_t=0}^{k-k_x} \widetilde{S}(k_x, k_t) (x - x_j)^{k_x} (t - t^n)^{k_t},
\end{aligned}$$

In this way, the local time evolution in each space-time control volume Ω_j is achieved correspondingly.

2.4. The W-B numerical fluxes

In particular, the numerical viscosity term $\alpha \left(U_{\tau, j+\frac{1}{2}}^\pm(t) - U_{\tau, j+\frac{1}{2}}^\pm(t) \right)$ in (8) is essential for the convection dominated problems. However, the existence of the numerical viscosity

term in (8) will destroy the W-B property in the case of the steady state. In this study, to distinguish the final W-B ADER-DG method, the ADER-DG method (7) along with numerical fluxes in (8) is termed as the non-W-B ADER-DG method.

Herein, to maintain the still water steady state (3), the W-B fluxes are constructed using the hydrostatic reconstruction technique [70]. Actually, the main idea of the hydrostatic reconstruction is to help us redefine conservative variables at inter-cells based on the hydrostatic variables. The key objective is to make the values of conservative variables at inter-cells equal in the case of the steady state. In practice, a typical hydrostatic reconstruction procedure mainly includes three steps: a transformation from the hydrostatic variables (which are constant in a given steady state, i.e., $u = 0$ and $h + b = C$); a consistent reconstruction based on the transformation from the hydrostatic variables; and transformation back to conservative variables in turn. Before applying the hydrostatic reconstruction, some notations are defined as

$$\begin{aligned} h_\tau(x, t) &= \frac{H_\tau(x, t)}{\sigma_\tau(x)}, \\ (h + b)_\tau(x, t) &= \frac{H_\tau(x, t) + (\sigma b)_\tau(x)}{\sigma_\tau(x)}, \\ \overline{(h + b)}_{\tau, j}(t) &= \frac{\overline{(H_\tau(x, t) + (\sigma b)_\tau)_j}}{\overline{\sigma_{\tau, j}}}, \\ (h + b)_{\tau, j+\frac{1}{2}}^\pm(t) &= \frac{H_{\tau, j+\frac{1}{2}}^\pm(t) + (\sigma b)_{\tau, j+\frac{1}{2}}^\pm}{\sigma_{\tau, j+\frac{1}{2}}^\pm}. \end{aligned} \quad (13)$$

Herein, the specific implementation procedures of the hydrostatic reconstruction are given in the following: Firstly, the channel and the water depth at inter-cell are reset as follows

$$\begin{aligned} \sigma_{\tau, j+\frac{1}{2}}^* &= \min \left(\sigma_\tau \left(x_{j+\frac{1}{2}}^+ \right), \sigma_\tau \left(x_{j+\frac{1}{2}}^- \right) \right), \\ h_{\tau, j+\frac{1}{2}}^{*, \pm}(t) &= \max \left(0, (h + b)_{\tau, j+\frac{1}{2}}^\pm(t) - \max(b_{\tau, j+\frac{1}{2}}^+, b_{\tau, j+\frac{1}{2}}^-) \right). \end{aligned} \quad (14)$$

Secondly, conservative variables at inter-cells are redefined by

$$U_{\tau, j+\frac{1}{2}}^{*, \pm}(t) = \begin{pmatrix} \sigma_{\tau, j+\frac{1}{2}}^* h_{\tau, j+\frac{1}{2}}^{*, \pm}(t) \\ \sigma_{\tau, j+\frac{1}{2}}^* h_{\tau, j+\frac{1}{2}}^{*, \pm}(t) u_\tau \left(x_{j+\frac{1}{2}}^\pm, t \right) \end{pmatrix}. \quad (15)$$

Finally, based on the hydrostatic reconstruction technique [70], the W-B numerical fluxes are constructed as follows

$$\begin{aligned}
\widehat{F}_{j+\frac{1}{2}}^- &= \frac{1}{2} \int_{t^n}^{t^{n+1}} \left[F \left(U_{\tau,j+\frac{1}{2}}^{*, -}(t), \sigma_{\tau,j+\frac{1}{2}}^* \right) + F \left(U_{\tau,j+\frac{1}{2}}^{*, +}(t), \sigma_{\tau,j+\frac{1}{2}}^* \right) - \alpha \left(U_{\tau,j+\frac{1}{2}}^{*, +}(t) - U_{\tau,j+\frac{1}{2}}^{*, -}(t) \right) \right] dt \\
&\quad + \left(\int_{t^n}^{t^{n+1}} \frac{1}{2} g \sigma_{\tau,j+\frac{1}{2}}^- \left(h_{\tau} \left(x_{j+\frac{1}{2}}^-, t \right) \right)^2 dt \right) - \left(\int_{t^n}^{t^{n+1}} \frac{1}{2} g \sigma_{\tau,j+\frac{1}{2}}^* \left(h_{\tau} \left(x_{j+\frac{1}{2}}^{*, -}, t \right) \right)^2 dt \right), \\
\widehat{F}_{j-\frac{1}{2}}^+ &= \frac{1}{2} \int_{t^n}^{t^{n+1}} \left[F \left(U_{\tau,j-\frac{1}{2}}^{*, -}(t), \sigma_{\tau,j-\frac{1}{2}}^* \right) + F \left(U_{\tau,j-\frac{1}{2}}^{*, +}(t), \sigma_{\tau,j-\frac{1}{2}}^* \right) - \alpha \left(U_{\tau,j-\frac{1}{2}}^{*, +}(t) - U_{\tau,j-\frac{1}{2}}^{*, -}(t) \right) \right] dt \\
&\quad + \left(\int_{t^n}^{t^{n+1}} \frac{1}{2} g \sigma_{\tau,j-\frac{1}{2}}^+ \left(h_{\tau} \left(x_{j-\frac{1}{2}}^+, t \right) \right)^2 dt \right) - \left(\int_{t^n}^{t^{n+1}} \frac{1}{2} g \sigma_{\tau,j-\frac{1}{2}}^* \left(h_{\tau} \left(x_{j-\frac{1}{2}}^{*, +}, t \right) \right)^2 dt \right).
\end{aligned} \tag{16}$$

Hereafter, the ADER-DG method (7) along with the numerical fluxes in (16) is referred to as the W-B ADER-DG method.

2.5. The source term approximation

Then, the discretization to the source term

$$\iint_{\Omega_j} S_{\tau}(x, t) v(x) dx dt \tag{17}$$

in (7) can be achieved using precise calculations such as quadrature rules, because $S_{\tau}(x, t)$ is a Taylor polynomial with the coefficients obtained by the DT procedure in section 2.3.

Similarly, other integrals

$$\int_{I_j} U_{\tau}(x, t^n) v(x) dx \quad \text{and} \quad \iint_{\Omega_j} F(U_{\tau}(x, t), \sigma_{\tau}(x)) v_x(x) dx dt. \tag{18}$$

in (7) can also be calculated by means of precise calculations. Thus, the ADER-DG method (7) for the shallow water flows in open channels (6) is completed and the proposed method is one-step, one-stage, fully-discrete.

2.6. The slope limiter

In addition, the solutions of the system (1) generally contain discontinuities and a slope limiter is imperative to compress possible non-physical oscillations near strong discontinuities. In this study, the classical total variation bounded (TVB) limiter [72, 73] is applied at time t^{n+1} .

Firstly, the TVB limiter needs to detect the troubled-cell (a cell contains discontinuities) according to the cell averages $\bar{U}_{\tau,j}(t^{n+1})$ as well as $\bar{U}_{\tau,j\pm 1}(t^{n+1})$ and inter-cell values $U_{\tau,j+\frac{1}{2}}^{\pm}(t^{n+1})$. To begin with, some notations are defined as follows

$$\begin{aligned}\tilde{U}_j &= U_{\tau,j+\frac{1}{2}}^-(t^{n+1}) - \bar{U}_{\tau,j}(t^{n+1}), \\ \tilde{\tilde{U}}_j &= \bar{U}_{\tau,j}(t^{n+1}) - U_{\tau,j-\frac{1}{2}}^+(t^{n+1}), \\ \Delta_+ \bar{U}_j &= \bar{U}_{\tau,j+1}(t^{n+1}) - \bar{U}_{\tau,j}(t^{n+1}), \\ \Delta_- \bar{U}_j &= \bar{U}_{\tau,j}(t^{n+1}) - \bar{U}_{\tau,j-1}(t^{n+1}).\end{aligned}\tag{19}$$

Then, the following updated values

$$\begin{aligned}\tilde{U}_j^{(\text{mod})} &= m(\tilde{U}_j, \Delta_+ \bar{U}_j, \Delta_- \bar{U}_j), \\ \tilde{\tilde{U}}_j^{(\text{mod})} &= m(\tilde{\tilde{U}}_j, \Delta_+ \bar{U}_j, \Delta_- \bar{U}_j),\end{aligned}\tag{20}$$

are obtained using the minmod function $m(\cdot, \cdot, \cdot)$, which is defined as

$$m(a_1, a_2, a_3) = \begin{cases} a_1, & \text{if } |a_1| \leq M\Delta x^2, \\ s \cdot \min_{1 \leq i \leq 3} |a_i|, & \text{if } |a_1| > M\Delta x^2 \text{ and } \text{sign}(a_1) = \text{sign}(a_2) = \text{sign}(a_3) = s, \\ 0, & \text{otherwise,} \end{cases}$$

with $M > 0$. If the updated values in (20) have changed from the original values in (19), i.e.,

$$\tilde{U}_j^{(\text{mod})} \neq \tilde{U}_j \quad \text{or} \quad \tilde{\tilde{U}}_j^{(\text{mod})} \neq \tilde{\tilde{U}}_j,$$

the cell I_j is then identified as a troubled-cell.

Subsequently, the limited inter-cell values $U_{\tau,j+\frac{1}{2}}^{\pm}(t^{n+1})$ are as follows

$$\begin{aligned}U_{\tau}^{(\text{mod})}\left(x_{j+\frac{1}{2}}^-, t^{n+1}\right) &= \bar{U}_j + \tilde{U}_j^{(\text{mod})}, \\ U_{\tau}^{(\text{mod})}\left(x_{j-\frac{1}{2}}^+, t^{n+1}\right) &= \bar{U}_j - \tilde{\tilde{U}}_j^{(\text{mod})}.\end{aligned}\tag{21}$$

Finally, a modified polynomial $U_\tau(x, t^{n+1})$ is obtained according to the limited inter-cell values (21) as well as the cell average $\bar{U}_{\tau,j}(t^{n+1})$.

However, the standard implementations of the TVB limiter will destroy the steady state (3). Therefore, to preserve the steady state (3), the first step of the TVB limiter here is to implement based on the inter-cell values and the cell averages of the free surface level $h + b$, namely

$$(h + b)_{\tau,j+\frac{1}{2}}^\pm(t^{n+1}) \quad \text{and} \quad \overline{(h + b)}_{\tau,j+k}(t^{n+1}), \quad \text{for } k = -1, 0, 1.$$

The modified polynomial of surface level $(h + b)_{\tau,j}^{(\text{mod})}(x, t^{n+1})$ can be obtained according to the above construction step as in (21). Then, the water depth and the wet cross section are obtained as follows

$$\begin{aligned} h_{\tau,j}^{(\text{mod})}(x, t^{n+1}) &= (h + b)_{\tau,j}^{(\text{mod})}(x, t^{n+1}) - b_{\tau,j}(x), \\ H_{\tau,j}^{(\text{mod})}(x, t^{n+1}) &= \sigma_{\tau,j}(x) h_{\tau,j}^{(\text{mod})}(x, t^{n+1}). \end{aligned}$$

At last, the modified mass flow rate $Q_{\tau,j}^{(\text{mod})}(x, t^{n+1})$ is obtained using the coefficients of $H_{\tau,j}^{(\text{mod})}(x, t^{n+1})$. In the case of the still water steady state (3), the cell averages and the inter-cell values of the surface level are all constants, which lead to

$$(h + b)_{\tau,j}^{(\text{mod})}(x, t^{n+1}) \equiv C, \quad \text{for } x \in I_j.$$

Consequently, there is no need to apply any slope limiter when the steady state is reached, and the W-B property is maintained accordingly.

2.7. The implementation steps of the ADER-DG method

For ease of applications, the detailed steps of the proposed method within one time step $t^n \rightarrow t^{n+1}$ are listed as follows.

- 1) Initially, to obtain $U_\tau(x, 0) = \sum_{k_x=0}^k \tilde{U}(k_x, 0)(x - x_j)^{k_x}$ in cell I_j , for $j = 1, 2, \dots, N$ by L^2 projection from the initial data $U(x, 0)$.
- 2) Using the recursive steps (11) and (12), to get $\tilde{U}(k_x, k_t)$, $\tilde{F}(k_x, k_t)$ and $\tilde{S}(k_x, k_t)$ on basis of $\tilde{U}(k_x, 0)$, and then to obtain $U_\tau(x, t)$, $F(U_\tau(x, t), \sigma_\tau(x))$ and $S_\tau(x, t)$ on each space-time control volumes Ω_j , for $j = 1, 2, \dots, N$.

- 3) Build W-B numerical fluxes $\hat{F}_{j-\frac{1}{2}}^+$ and $\hat{F}_{j+\frac{1}{2}}^-$ using the formula (16).
- 4) Discretize the integrals in (7) by precise calculations.
- 5) Update to the next time level $U_\tau(x, t^{n+1})$ in the light of the one-stage formula (7).
- 6) Employ the TVB slope limiter on $U_\tau(x, t^{n+1})$ when needed.
- 7) Repeat the steps 2)–6).

2.8. Summary of the W-B method

Proposition 1. *By means of the W-B numerical fluxes (16) and the numerical integrations of the source terms (17) as well as other terms (18), the ultimate ADER-DG method (7) is a W-B method for the still water steady state solutions (3). This implies that $U_\tau(x, t^{n+1}) = U_\tau(x, t^n)$.*

Proof. Firstly, according to Eq. (13) and the DT procedure in section 2.3, the following facts

$$u_\tau(x, t) = 0, \quad h_\tau(x, t) + b_\tau(x) = C,$$

are evident at the initial steady state (3), which signify that the local space-time polynomial representation of the solutions maintains the steady state in each space-time control volumes Ω_j , for $j = 1, 2, \dots, N$.

Obviously, the W-B property holds for the first equation of the system (1), due to the physical flux Q being zero since $u = 0$. Therefore, all attention should be focused on the second momentum equation of the system (1).

Actually, at the initial still water steady state (3), the following equalities

$$h_{\tau, j \pm \frac{1}{2}}^{*, -}(t) = h_{\tau, j \pm \frac{1}{2}}^{*, +}(t), \quad U_{\tau, j \pm \frac{1}{2}}^{*, -}(t) = U_{\tau, j \pm \frac{1}{2}}^{*, +}(t),$$

are realized according to the constructed conservative variable (15) as well as the hydrostatic reconstruction for the water depth (14). At this point, the original W-B fluxes (16) reduce

to the following simple forms

$$\begin{aligned}\widehat{F}_{j-\frac{1}{2}}^+ &= \begin{pmatrix} 0 \\ \int_{t^n}^{t^{n+1}} \frac{1}{2} g \sigma_{\tau, j-\frac{1}{2}}^+ \left(h_{\tau} \left(x_{j-\frac{1}{2}}^+, t \right) \right)^2 dt \end{pmatrix} = \int_{t^n}^{t^{n+1}} F \left(U_{\tau} \left(x_{j-\frac{1}{2}}^+, t \right), \sigma_{\tau} \left(x_{j-\frac{1}{2}}^+ \right) \right) dt, \\ \widehat{F}_{j+\frac{1}{2}}^- &= \begin{pmatrix} 0 \\ \int_{t^n}^{t^{n+1}} \frac{1}{2} g \sigma_{\tau, j+\frac{1}{2}}^- \left(h_{\tau} \left(x_{j+\frac{1}{2}}^-, t \right) \right)^2 dt \end{pmatrix} = \int_{t^n}^{t^{n+1}} F \left(U_{\tau} \left(x_{j+\frac{1}{2}}^-, t \right), \sigma_{\tau} \left(x_{j+\frac{1}{2}}^- \right) \right) dt.\end{aligned}$$

Therefore, the error between the approximations to the flux gradient and to the source term of (7) in the the second momentum equation writes as follows

$$\begin{aligned}R &= \widehat{F}_{j+\frac{1}{2}}^- \cdot v \left(x_{j+\frac{1}{2}}^- \right) - \widehat{F}_{j-\frac{1}{2}}^+ \cdot v \left(x_{j-\frac{1}{2}}^+ \right) \\ &\quad - \iint_{\Omega_j} F \left(U_{\tau}(x, t), \sigma_{\tau}(x) \right) v_x dx dt - \iint_{\Omega_j} S_{\tau}(x, t) v_x dx dt \\ &= \int_{t^n}^{t^{n+1}} F \left(U_{\tau} \left(x_{j+\frac{1}{2}}^-, t \right), \sigma_{\tau} \left(x_{j+\frac{1}{2}}^- \right) \right) dt \cdot v \left(x_{j+\frac{1}{2}}^- \right) \\ &\quad - \int_{t^n}^{t^{n+1}} F \left(U_{\tau} \left(x_{j-\frac{1}{2}}^+, t \right), \sigma_{\tau} \left(x_{j-\frac{1}{2}}^+ \right) \right) dt \cdot v \left(x_{j-\frac{1}{2}}^+ \right) \\ &\quad - \iint_{\Omega_j} F \left(U_{\tau}(x, t), \sigma_{\tau}(x) \right) v_x dx dt - \iint_{\Omega_j} S_{\tau}(x, t) v_x dx dt \\ &= \iint_{\Omega_j} \frac{\partial F \left(U_{\tau}(x, t), \sigma_{\tau}(x) \right)}{\partial x} v(x) dx dt - \iint_{\Omega_j} S_{\tau}(x, t) v_x dx dt \\ &= \iint_{\Omega_j} \left(\frac{\partial F \left(U_{\tau}(x, t), \sigma_{\tau}(x) \right)}{\partial x} - S_{\tau}(x, t) \right) v_x dx dt \\ &= 0, \quad \text{for } j = 1, 2, \dots, N.\end{aligned} \tag{22}$$

The last equality in (22) stems from the following derivations

$$\begin{aligned}\left(\frac{1}{2} g \left(\frac{H^2}{\sigma} \right)_{\tau} \right)_x &= \left(\frac{1}{2} g \sigma_{\tau} h_{\tau}^2 \right)_x \\ &= \frac{1}{2} g h_{\tau}^2 (\sigma_{\tau})_x + \frac{1}{2} g \sigma_{\tau} (h_{\tau} + b_{\tau} - b_{\tau})_x^2 \\ &= \frac{1}{2} g h_{\tau}^2 (\sigma_{\tau})_x + \frac{1}{2} g \sigma_{\tau} (C^2)_x - g \sigma_{\tau} (h_{\tau} + b_{\tau}) (b_{\tau}(x))_x + g \sigma_{\tau} b_{\tau} (b_{\tau})_x \\ &= \frac{1}{2} g h_{\tau}^2 (\sigma_{\tau})_x - g \sigma_{\tau} h_{\tau} (b_{\tau})_x \\ &= -\frac{1}{2} g H_{\tau}^2 \left(\frac{1}{\sigma_{\tau}} \right)_x - g H_{\tau} (b_{\tau})_x,\end{aligned}$$

which means

$$\frac{\partial F(U_\tau(x, t), \sigma_\tau(x))}{\partial x} = S_\tau(x, t)$$

in (22) at the still water steady state. The observation in (22) signifies that

$$U_\tau(x, t^{n+1}) = U_\tau(x, t^n), \quad x \in I_j, \text{ for } j = 1, 2, \dots, N$$

from (7) due to the arbitrariness of the test function $v(x) \in V_\tau^k$. This means that the current method is W-B and maintains the still water steady state solutions. Finally, the proof is completed accordingly. $\square$

3. Numerical results

This section implements extensive examples to validate the current third-order (i.e., $k = 2$ in space V_τ^k) ADER-DG method along with $g = 9.812$ and the CFL number as 0.18.

3.1. Accuracy test

The first example is borrowed from [38] and is used here to testify the accuracy order. The bottom topography, channel width, and initial data are defined by

$$\begin{aligned} b(x) &= \sin^2(\pi x), \\ \sigma(x) &= e^{\sin(2\pi x)}, \\ H(x, 0) &= \sigma(x) (5 + e^{\cos(2\pi x)}), \\ Q(x, 0) &= \sigma(x) \sin(\cos(2\pi x)), \end{aligned} \quad x \in [0, 1].$$

Then, this example is numerically solved up to $T = 0.1$. Table 2 lists the L^1 errors and the orders for H and Q . It is evident that the third-order accuracy is achieved as expected.

3.2. W-B test

This example from [21, 25] is used to numerically verify the W-B property on account of the below bottom

$$b(x) = \begin{cases} 0.25(1 + \cos(10\pi(x - 0.5))), & \text{if } 0.4 \leq x \leq 0.6, \\ 0, & \text{otherwise,} \end{cases} \quad x \in [0, 1]. \quad (23)$$

Table 2: L^1 errors and orders of H and Q in section 3.1.

Cells	H		Q	
	L^1 error	Order	L^1 error	Order
25	1.22E-04		1.68E-04	
50	2.04E-05	2.58	2.75E-05	2.61
100	3.30E-06	2.63	4.23E-06	2.70
200	4.80E-07	2.78	6.30E-07	2.75
400	6.66E-08	2.85	8.61E-08	2.87
800	8.38E-09	2.99	1.09E-09	2.98
1600	1.03E-09	3.02	1.35E-09	3.01

The channel with varying width $\sigma(x)$ has the below form

$$\sigma(x) = \begin{cases} 1 - \sigma_0 \left(1 + \cos \left(2\pi \frac{x - (x_l + x_r)/2}{x_r - x_l} \right) \right), & \text{if } x \in [x_l, x_r], \\ 1, & \text{otherwise,} \end{cases} \quad (24)$$

where x_l and x_r denote the left and right boundary of the contraction, and $1 - 2\sigma_0$ is the minimum width of the channel at point $(x_l + x_r)/2$. Herein, $x_l = 0.25$, $x_r = 0.75$ and $\sigma_0 = 0.2$ are used in this example. 3D views of the bottom b and the channel terrain σ are shown in Fig. 1. The initial data is specified as still steady states

$$H = \sigma(x) [1 - b(x)], \quad Q = 0. \quad (25)$$

Fig. 2 shows the numerical surface level $h + b$ at $T = 1$ on a mesh with 200 cells, the bottom topography b , and the channel terrain σ . The errors of H and Q are presented in Table 3. The errors are all at the same level of the machine precision, and the expected W-B property is verified accordingly.

3.3. Small perturbation test

This example [21] is applied here to test the ability of the current method to capture small perturbations. A small perturbation is imposed to the same still water steady states

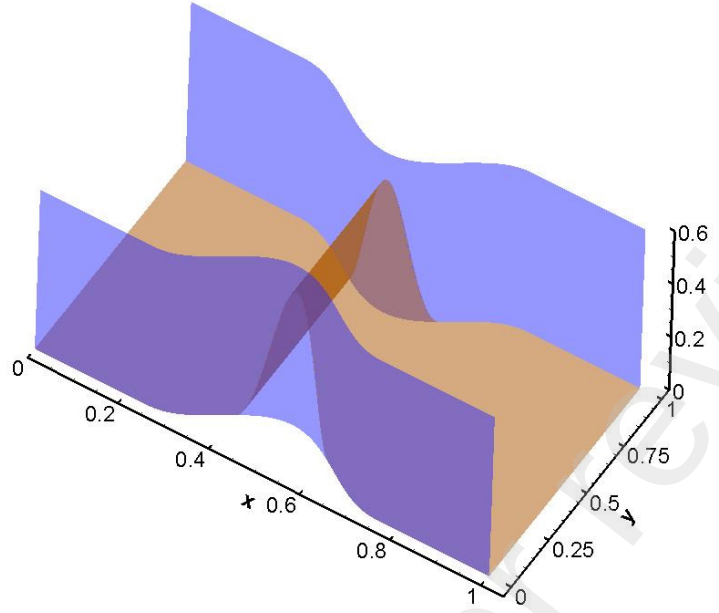


Fig. 1: The 3D view of the bottom b and the channel terrain σ of section 3.2.

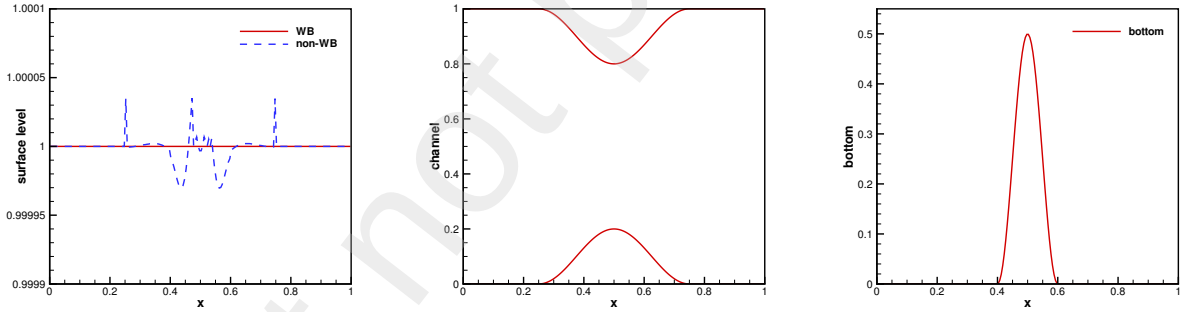


Fig. 2: The still water steady state solutions in section 3.2. Left: the surface level $h + b$; Middle: the shape of the channel σ ; Right: the bottom topography b .

Table 3: L^1 , L^2 and L^∞ errors of the still water steady state solutions in section 3.2.

Precision	L^1 error		L^2 error		L^∞ error	
	H	Q	H	Q	H	Q
Single	2.25E-07	1.42E-07	2.29E-07	5.14E-07	7.36E-07	6.23E-07
Double	3.61E-14	4.54E-14	3.67E-14	6.31E-14	5.72E-14	4.56E-14

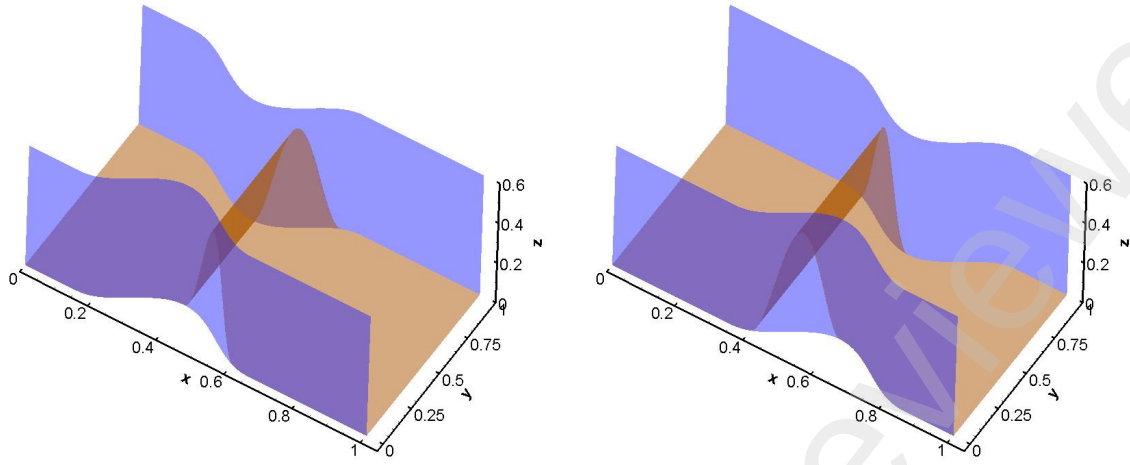


Fig. 3: The 3D views of two types of channels. Left: the channel with a left shifted contraction; Right: the channel with a right shifted contraction.

(23)–(25) and enjoys the below form

$$H(x, 0) = \begin{cases} \sigma(x) (1.01 - b(x)), & \text{if } 0.1 \leq x \leq 0.2, \\ 1, & \text{otherwise,} \end{cases} \quad Q = 0.$$

Two types of channels $\sigma(x)$ in (24) are applied: one with a left shifted contraction $x_l = 0.15$, $x_r = 0.65$, $\sigma_0 = 0.2$, and another with a right shifted contraction $x_l = 0.35$, $x_r = 0.85$, $\sigma_0 = 0.2$. 3D views of the channel and the vertical profile of the bottom are shown in Fig. 3.

Fig. 4 presents numerical results by the W-B method against those by the non-W-B one on the same mesh with 200 cells at different end time. Obviously, the numerical results obtained by the W-B method fit well with the reference ones and can successfully capture the small perturbations even on a relatively coarse mesh. However, the non-W-B method leads to non-physical oscillations compared with those by the W-B one.

3.4. A converging-diverging channel

Further, this example [25, 74] is considered over a flat bottom (i.e., $b = 0$) and with the following converging-diverging channel

$$\sigma(x) = \begin{cases} 5 - 0.7065 \left(1 + \cos \left(2\pi \frac{x - 250}{300} \right) \right), & \text{if } x \in [150, 450], \\ 5, & \text{otherwise,} \end{cases} \quad x \in [0, 500].$$

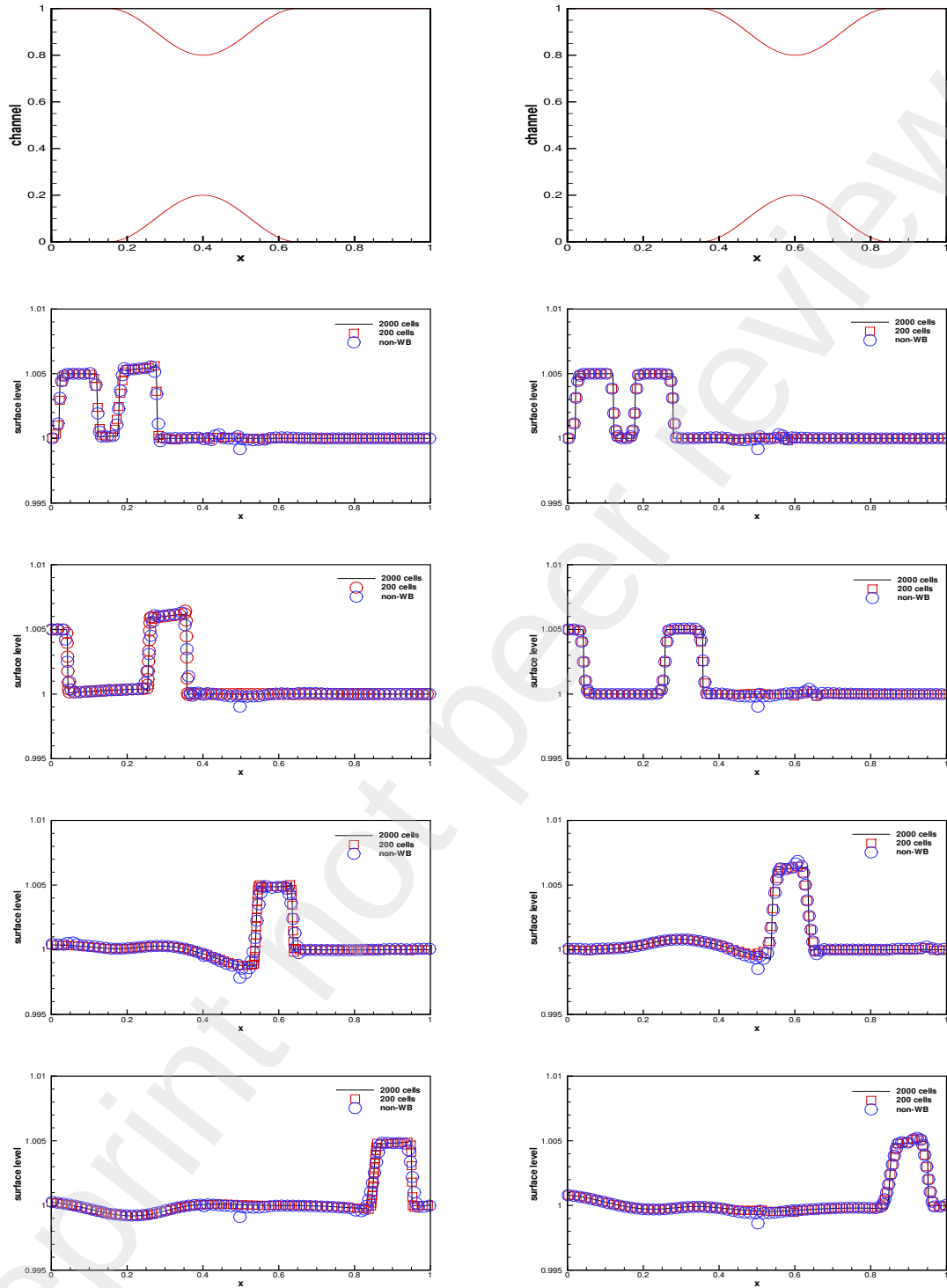


Fig. 4: The contracting channel $\sigma(x)$ (the top row) and numerical results by the W-B method against those by the non-W-B one at $T = 0.025, 0.05, 0.15$ and 0.25 (from the second row to the bottom) in section 3.3. Left: the channel with a left shifted contraction; Right: the channel with a right shifted contraction.

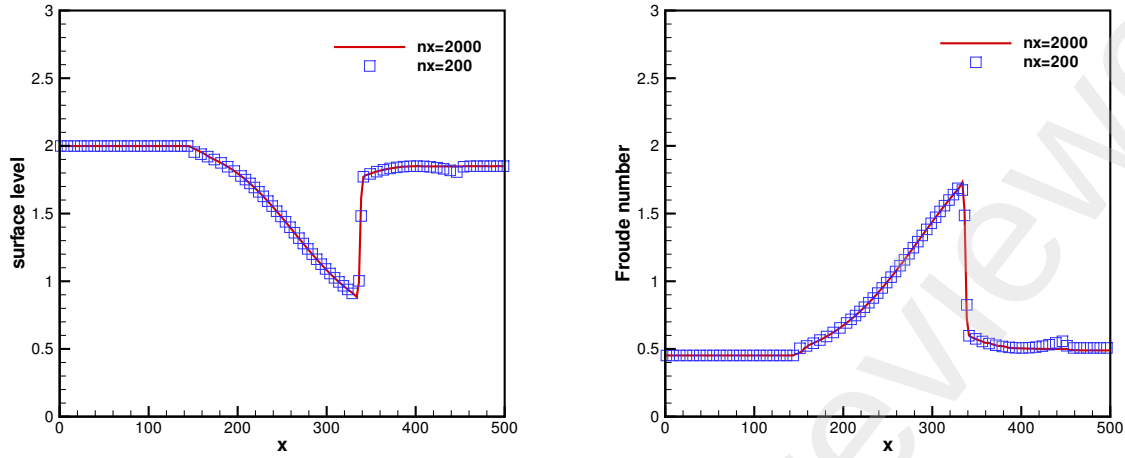


Fig. 5: Steady transcritical flow in a converging-diverging channel of section 3.4. Left: the surface level $h+b$; Right: the Froude number.

The initial data are

$$H(x, 0) = 2\sigma(x), \quad Q(x, 0) = 20.$$

The boundary conditions are imposed by $Q = 20$ at the upstream and $H = 9.25$ at the downstream. The water height h and the Froude numbers at $T = 5000$ (until it reaches a steady state) on meshes with 200 and 2000 uniform cells are shown in Fig. 5. Obviously, the numerical results can be comparable with those in [1, 74].

3.5. Moving water steady states over a hump

In the end, this example [25] is used to validate the performance of the proposed method when the shallow water flows converge to the moving water steady states over a bump bottom along with different channel configurations. The bottom consists of a bump

$$b(x) = \begin{cases} 0.2 - 0.05(x - 10)^2, & \text{if } 8 \leq x \leq 12, \\ 0, & \text{otherwise,} \end{cases} \quad x \in [0, 25]$$

and with the same channel width $\sigma(x)$ as in (24). The initial data are

$$H(x, 0) = \sigma(x) [0.5 - b(x)], \quad Q(x, 0) = 0.$$

This example is numerically solved up to $T = 200$ on a mesh with 200 uniform cells, when the solutions reach the moving water steady state. Additionally, two types of variable channel breadths and boundary conditions are considered to testify the effects on the final steady flows, which can be subcritical or transcritical with or without a steady shock.

(a) Subcritical flow.

The boundary condition is imposed by $Q = 4.42$ at the upstream and $H = 2$ at the downstream. Herein, two sets of channel breadths are handled: one with a left shifted contraction $x_l = 3.75$, $x_r = 16.25$, $\sigma_0 = 0.05$ and the other with a right shifted contraction $x_l = 8.75$, $x_r = 21.25$, $\sigma_0 = 0.05$. The surface level $h + b$, the mass flow rate Q at $T = 200$, and 3D views of the bottom b , channel shape and water surface are shown in Fig. 6. For comparison, the analytical solutions are also presented in Fig. 6. Clearly, the converged state is a subcritical flow and the numerical solutions are in good agreement with the analytic ones.

(b) Transcritical flow without a shock.

The boundary condition is imposed by $Q = 1.53$ at the upstream and $H = 0.66$ at the downstream. Herein, two sets of channels are considered: one with a left shifted contraction $x_l = 3.75$, $x_r = 16.25$, $\sigma_0 = 0.15$ and the other with a right shifted contraction $x_l = 8.75$, $x_r = 21.25$, $\sigma_0 = 0.15$. Fig. 7 illustrates the surface level $h + b$, the mass flow rate Q at $T = 200$ along with the 3D views of the bottom b , channel shape and water surface. It is obvious that the converged steady state is a transcritical flow free of a shock, and the numerical results are in good agreement with the analytical ones, see Fig. 7.

(c) Transcritical flow with a shock.

The boundary condition is defined as $Q = 0.18$ at the upstream and $H = 0.33$ at the downstream. Herein, we consider two sets of channels, one with a left shifted contraction $x_l = 3.75$, $x_r = 16.25$, $\sigma_0 = 0.15$, and the other with a right shifted contraction $x_l = 8.75$, $x_r = 21.25$, $\sigma_0 = 0.15$. Fig. 8 presents the surface level $h + b$, the mass flow rate Q at $T = 200$, and 3D views of the bottom b , channel shape and water surface. To have a comparison, the analytical solutions are also shown in Fig. 8. Obviously, the converged state is a transcritical flow with a shock appearing in the middle of the domain, and the numerical results fit well

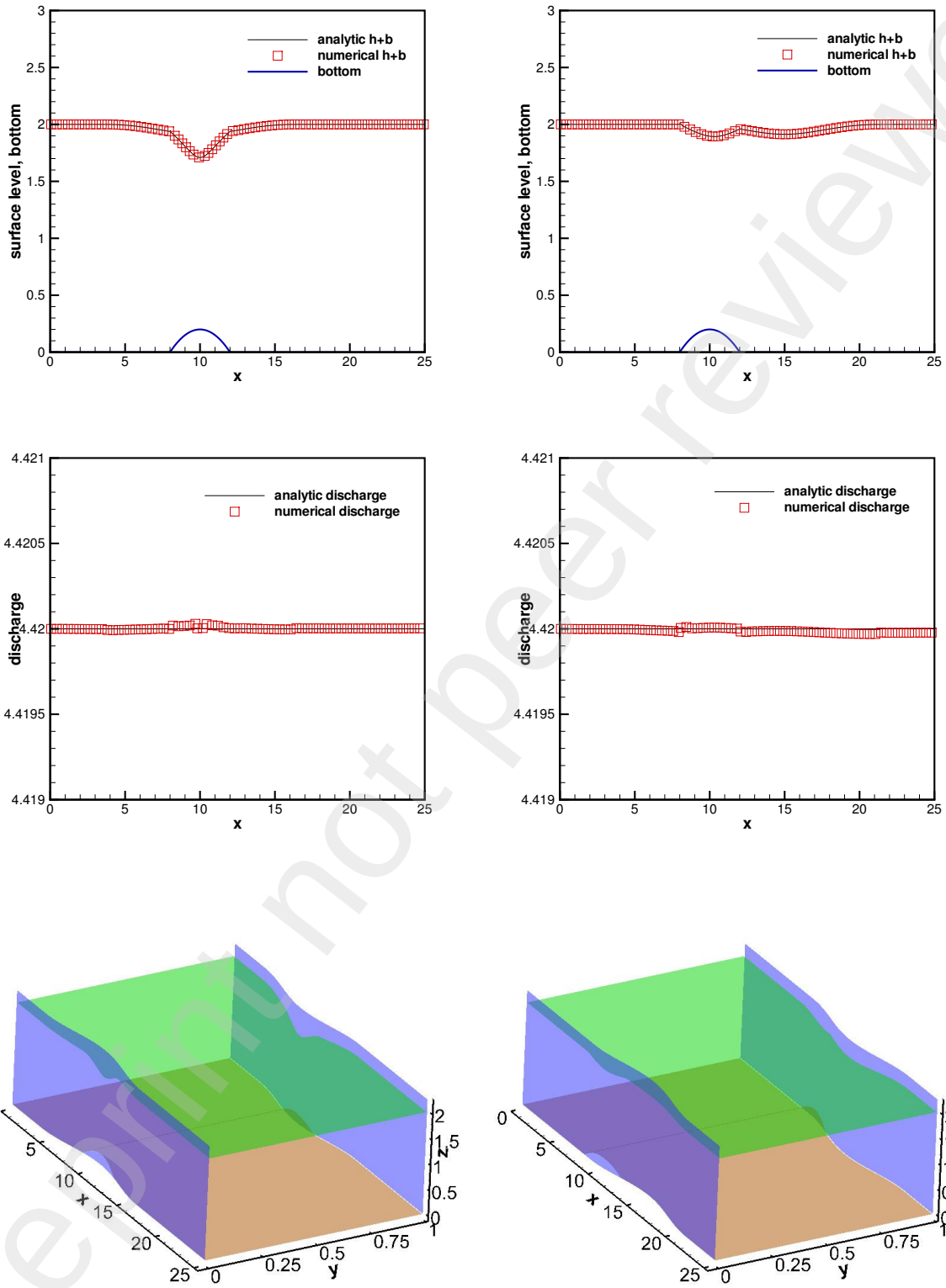


Fig. 6: Steady subcritical flow over a bump in section 3.5. Top: the surface level $h+b$; Middle: the discharge Q ; Bottom: the 3D views of the bottom b , water surface $h+b$ and channel shape σ . Left: the channel with a left shifted contraction; Right: the channel with a right shifted contraction.

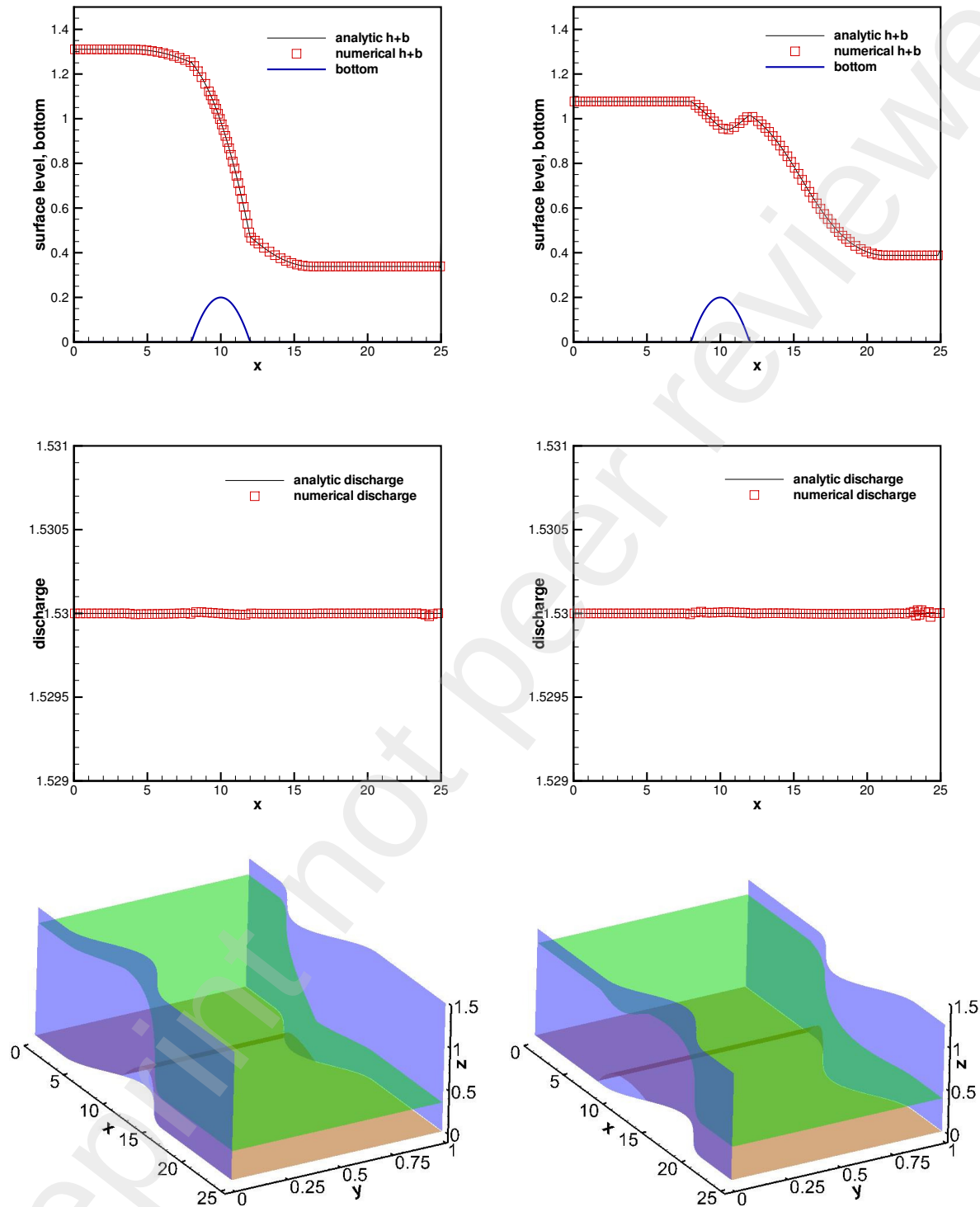


Fig. 7: Steady subcritical flow over a bump without a shock in section 3.5. Top: the surface level $h + b$; Middle: the discharge Q ; Bottom: the 3D views of the bottom topography b , water surface $h + b$ and channel shape σ . Left: the channel with a left shifted contraction; Right: the channel with a right shifted contraction.

with the analytical ones.

4. Conclusions

This article presents a new high-order W-B ADER-DG method for the shallow water flows in open channels over uneven bottom topography. Different from the traditional RKDG methods, which apply the multi-stage Runge-Kutta approach, the method here applies the ADER approach to achieve the temporal discretization. To realize the local temporal evolution in each space-time control volume, the DT procedure is used here to obtain spatiotemporal expansion coefficients through the spatial ones. The DT procedure is more efficient than the C-K procedure from the traditional ADER methods and the coding of the DT procedure is more concise. In brief, the current method is one-step, one-stage, fully-discrete and can theoretically gain arbitrary high-order accuracy in both space and time. In addition, the usage of the DT procedure makes the resulting method to avoid solving generalized Riemann problems at inter-cells. The W-B numerical fluxes constructed using the hydrostatic reconstruction technique along with simple source term discretizations lead to the expected W-B property accordingly. A number of numerical examples strongly suggest that the proposed method maintains the W-B property, keeps genuine high-order accuracy, and accurately captures small perturbations to the still water steady state.

5. Acknowledgments

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Appendix A. The algorithm of the DT procedure for the shallow water flows in open channels

Input: $\tilde{U}(k_x, 0, 1 : 2)$, $\tilde{\sigma}(k_x)$, and $\tilde{b}(k_x)$

Output: $\tilde{U}(k_x, k_t, 1 : 2)$, $\tilde{F}(k_x, k_t, 1 : 2)$ and $\tilde{S}(k_x, k_t)$,

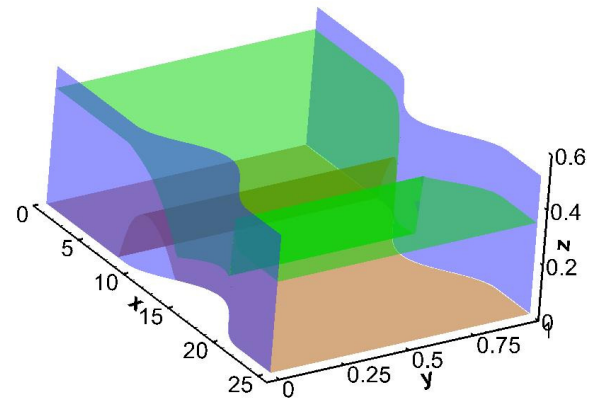
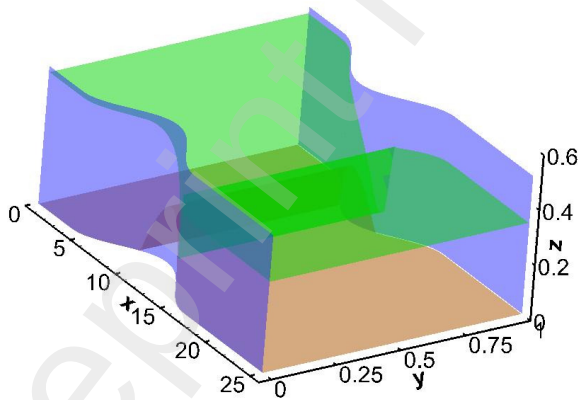
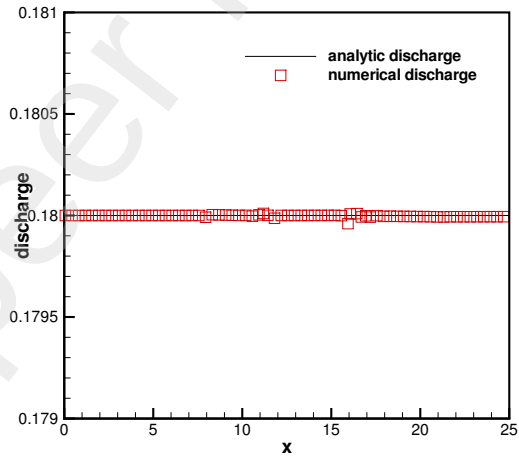
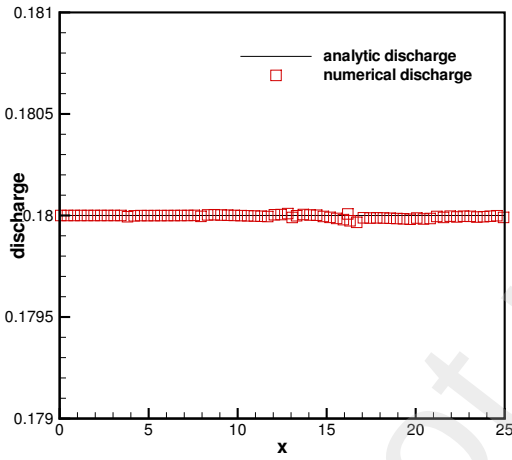
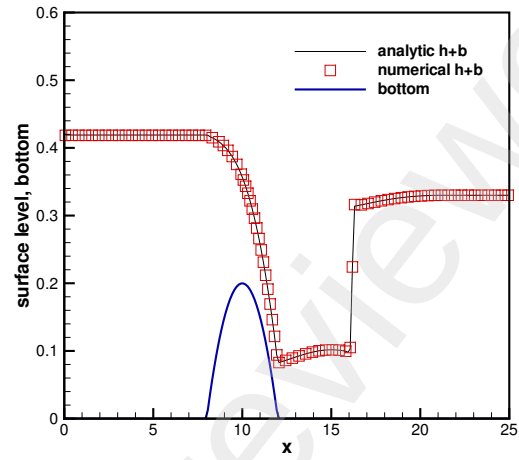
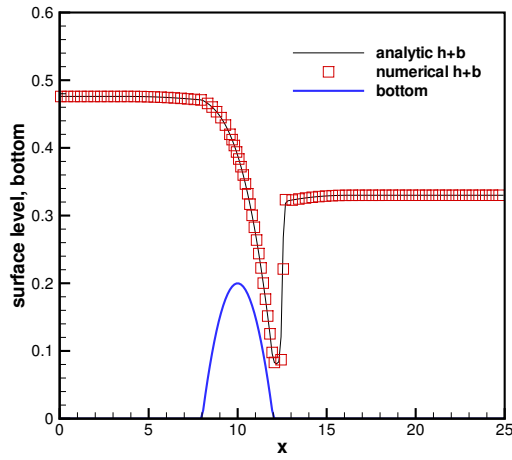


Fig. 8: Steady subcritical flow over a bump with a shock in section 3.5. Top: the surface level $h+b$; Middle: the discharge Q ; Bottom: the 3D views of the bottom b , water surface level $h+b$ and channel shape σ . Left: the channel with a left shifted contraction; Right: the channel with a right shifted contraction.

$$\begin{aligned}
\% \quad \widetilde{U}(k_x, k_t, 1) &\triangleq \widetilde{H}(k_x, k_t), \quad \widetilde{U}(k_x, k_t, 2) \triangleq \widetilde{Q}(k_x, k_t) \\
\% \quad \widetilde{F}(k_x, k_t, 1) &\triangleq \widetilde{Q}(k_x, k_t), \quad \widetilde{F}(k_x, k_t, 2) \triangleq \widetilde{G}_1(k_x, k_t) + \frac{1}{2} * g * \widetilde{G}_2(k_x, k_t) \\
\% \quad \widetilde{S}(k_x, k_t) &\triangleq -\frac{1}{2} * \widetilde{G}_3(k_x, k_t) - g * \widetilde{G}_4(k_x, k_t) \\
\% \quad \widetilde{G}_1(k_x, k_t) &\triangleq \frac{\widetilde{Q}^2}{H}(k_x, k_t), \quad \widetilde{G}_2(k_x, k_t) \triangleq \frac{\widetilde{H}^2}{\sigma}(k_x, k_t) \\
\% \quad \widetilde{G}_3(k_x, k_t) &\triangleq \widetilde{H^2 \left(\frac{1}{\sigma} \right)_x}(k_x, k_t), \quad \widetilde{G}_4(k_x, k_t) \triangleq \widetilde{Hb_x}(k_x, k_t)
\end{aligned}$$

while cell I_j , $j = 1, N$ **do**

$$\widetilde{\sigma_inv}(0) = 1.0/\widetilde{\sigma}(0) \quad \% \text{ “}\sigma_inv\text{” denotes “}\frac{1}{\sigma}\text{”}$$

$$tot_ \sigma_inv = 0$$

while $r_x = 0$, k_x **do**

$$tot_ \sigma_inv = tot_ \sigma_inv + \widetilde{\sigma}(r_x) * \widetilde{\sigma_inv}(k_x - r_x)$$

end while

if $k_x > 0$ **then**

$$\widetilde{\sigma_inv}(k_x) = -tot_ \sigma_inv/\widetilde{\sigma}(0)$$

end if

while $k_x = 0$, k **do**

$$\widetilde{b_x}(k_x) = (k_x + 1)\widetilde{b}(k_x + 1) \quad \% \text{ “}b_x\text{” denotes “}b_x\text{”}$$

$$\widetilde{\sigma_inv_x}(k_x) = (k_x + 1)\widetilde{\sigma_inv}(k_x + 1) \quad \% \text{ “}\sigma_inv_x\text{” denotes “}\left(\frac{1}{\sigma}\right)_x\text{”}$$

end while

$$\widetilde{H_inv}(0, 0) = 1.0/\widetilde{U}(0, 0, 1) \quad \% \text{ “}H_inv\text{” denotes “}\frac{1}{H}\text{”}$$

while $k_t = 0, k$ **do**

while $k_x = 0$, $k - k_t$ **do**

$$tot_H2 = 0 \quad \% \text{ “}H2\text{” denotes “}H^2\text{”}$$

$$tot_Q2 = 0 \quad \% \text{ “}Q2\text{” denotes “}Q^2\text{”}$$

$$tot_H_inv = 0$$

while $r_t = 0$, k_t **do**

while $r_x = 0$, k_x **do**

$$tot_H2 = tot_H2 + \widetilde{U}(r_x, r_t, 1) * \widetilde{U}(k_x - r_x, k_t - r_t, 1)$$

$$tot_Q2 = tot_Q2 + \widetilde{U}(r_x, r_t, 2) * \widetilde{U}(k_x - r_x, k_t - r_t, 2)$$

$$tot_H_inv = tot_H_inv + \widetilde{U}(r_x, r_t, 1) * \widetilde{H_inv}(k_x - r_x, k_t - r_t)$$

end while

end while

$$\widetilde{H2}(k_x, k_t) = tot_H2$$

$$\widetilde{Q2}(k_x, k_t) = tot_Q2$$

if $k_x + k_t > 0$ **then**

$$\widetilde{H_inv}(k_x, k_t) = -tot_H_inv / \widetilde{U}(0, 0, 1)$$

end if

$$tot_G_1 = 0 \quad \% \text{ “} G_1 \text{” denotes “} \frac{Q^2}{H} \text{”}$$

$$tot_G_2 = 0 \quad \% \text{ “} G_2 \text{” denotes “} \frac{H^2}{\sigma} \text{”}$$

$$tot_G_3 = 0 \quad \% \text{ “} G_3 \text{” denotes “} H^2 \left(\frac{1}{\sigma} \right)_x \text{”}$$

$$tot_G_4 = 0 \quad \% \text{ “} G_4 \text{” denotes “} Hb_x \text{”}$$

while $r_x = 0, k_x$ **do**

while $r_t = 0, k_t$ **do**

$$tot_G_1 = tot_G_1 + \widetilde{Q2}(r_x, r_t) * \widetilde{H_inv}(k_x - r_x, k_t - r_t)$$

end while

$$tot_G_2 = tot_G_2 + \widetilde{H2}(r_x, k_t) * \widetilde{\sigma_inv}(k_x - r_x)$$

$$tot_G_3 = tot_G_3 + \widetilde{H2}(r_x, k_t) * \widetilde{\sigma_inv_x}(k_x - r_x)$$

$$tot_G_4 = tot_G_4 + \widetilde{H}(r_x, k_t) * \widetilde{b_x}(k_x - r_x)$$

end while

$$\widetilde{G}_1(k_x, k_t) = tot_G_1$$

$$\widetilde{G}_2(k_x, k_t) = tot_G_2$$

$$\widetilde{G}_3(k_x, k_t) = tot_G_3$$

$$\widetilde{G}_4(k_x, k_t) = tot_G_4$$

$$\widetilde{F}(k_x, k_t, 1) = \widetilde{U}(k_x, k_t, 2)$$

$$\widetilde{F}(k_x, k_t, 2) = \widetilde{G}_1(k_x, k_t) + \frac{1}{2} * g * \widetilde{G}_2(k_x, k_t)$$

end while

if $k_t < k$ **then**

```

while  $k_x = 0, k - (k_t + 1)$  do
     $\tilde{S}(k_x, k_t) = -\frac{1}{2} * \tilde{G}_3(k_x, k_t) - g * \tilde{G}_4(k_x, k_t)$     % the source term
     $\tilde{U}(k_x, k_t + 1, 1) = -\frac{k_x + 1}{k_t + 1} * \tilde{F}(k_x + 1, k_t, 1)$ 
     $\tilde{U}(k_x, k_t + 1, 2) = -\frac{k_x + 1}{k_t + 1} * \tilde{F}(k_x + 1, k_t, 2) + \frac{1}{k_t + 1} * \tilde{S}(k_x, k_t)$ 
end while
end if
end while
end while

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